

CONFIDENTIAL: CORE BANKING OPERATIONS & FUNCTIONAL WORKFLOW BLUEPRINT

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1. Executive Summary

This document outlines the operational heart of the banking institution. While the IT infrastructure provides the tools, the Core Banking Operations define the actual flow of capital, clearing and settlement mechanisms, daily accounting cycles, and regulatory treasury management. This blueprint maps how the bank functions on a day-to-day basis.

2. The Core Banking System (CBS) Operations

The CBS is the centralized ledger that ensures every branch and digital channel reflects the same truth. It operates on a strict double-entry accounting system.

General Ledger (GL) Management: Every transaction hits the GL. If a customer deposits money, the customer's account is credited (liability for the bank), and the bank's cash account is debited (asset for the bank).

Maker-Checker (Dual Control): For high-value transactions or system configuration changes, the operation must be initiated by one employee (Maker) and authorized by another independent employee (Checker) to prevent internal fraud.

Start-of-Day (SOD) & End-of-Day (EOD) Operations:

SOD: System wakes up, applies new interest rates, and opens posting channels.

EOD: The most critical batch process. The system goes offline for maintenance, calculates interest for millions of accounts, deducts monthly charges, reconciles ATM networks, and generates daily trial balances for regulatory reporting.

3. Transaction Flow: How Money Actually Moves

Money movement is categorized into internal transfers and inter-bank settlements.

A. Intra-Bank Transfers (Within the same bank)

Instant Ledger Updates: If Customer A sends money to Customer B (both in the same bank), the CBS simply debits A and credits B. No external money movement occurs.

B. Inter-Bank Transfers & Clearing (Sending to another bank)

Payment Networks (UPI, IMPS, NEFT, RTGS): When a user initiates a UPI or NEFT transfer, the bank's API gateway formats the message and sends it to the central clearing house.

Clearing & Settlement: The central authority records the transaction. At specific intervals (or in real-time for RTGS/UPI), the central bank settles the net differences between Bank A and Bank B's master accounts maintained with the central bank.

4. The Loan Lifecycle (Asset Operations)

Lending is the primary revenue source, and its operational flow is strictly governed.

Sourcing & Origination: Collecting customer data and KYC documents via branches or digital Loan Origination Systems (LOS).

Underwriting: Assessing the borrower's repayment capacity using credit scores, income analysis, and predictive ML models for risk intelligence.

Disbursement: Once approved, the funds are released into the borrower's account.

Collection & NPA Management: Tracking monthly EMIs. If a borrower defaults for 90 days, the asset is classified as a Non-Performing Asset (NPA), triggering legal and recovery operations.

5. Branch & Agency Operations

Despite digital banking, physical branches manage critical localized operations.

Cash Management: Branches maintain a specific vault limit. Excess cash is transported to centralized Currency Chests securely, while cash shortages are replenished via automated forecasting models.

Clearing House (Cheques): Physical cheques deposited at branches are scanned and sent through the Cheque Truncation System (CTS). The physical paper is destroyed, and the digital image is used for inter-bank clearing.

6. Treasury & Regulatory Operations

The bank cannot lend out every single rupee it receives. It must manage its own liquidity and comply with central bank mandates.

Cash Reserve Ratio (CRR): A strict percentage of total deposits that the bank must keep in liquid cash with the central bank (earning zero interest) to ensure liquidity.

Statutory Liquidity Ratio (SLR): A percentage of deposits that must be invested in safe, liquid assets like government bonds or gold.

Asset Liability Management (ALM): Ensuring the bank has enough cash coming in (from loan repayments) to match the cash going out (when depositors withdraw). Mismatches here cause bank failures.